

Yuan Zhang

(+61) 402 509 083 | dollaroo099@gmail.com |

Website: <https://yuanyuandollar.github.io/dollar-zhang-earth-online/>

PROFILE

Actuarial Studies master's candidate at UNSW with experience across insurance product design, pricing, claims and exposure analysis, and customer analytics. Strong quantitative foundation in actuarial modelling and data analysis, combined with a demonstrated ability to translate risk insights into practical product and commercial recommendations. Interested in building a career in commercial insurance and underwriting.

EDUCATION & QUALIFICATIONS

University of New South Wales | Master of Actuarial Studies (Extension) | Sep 2024 – Sep 2026 (Expected)

- Actuaries of Tomorrow Scholarship, IFoA Foundation (2024)
- Relevant studies: actuarial statistics and insurance risk modelling, finance, business economics, risk management and actuarial data analytics

Xi'an Jiaotong-Liverpool University | Bachelor of Actuarial Science | Sep 2020 – Jun 2024

- University Academic Achievement Award (2022–23)
- IFoA exemptions: CM1, CM2 and CS1

INSURANCE & COMMERCIAL PROJECTS

Actuaries in Space: The Pricing Frontier | SOA Research Institute Student Research Case Study Challenge | 2026

- Analysed historical claims and prospective exposure data across four hazard lines to model expected losses, volatility and tail risk for a new commercial insurance opportunity
- Developed short- and long-term pricing estimates and stress-tested portfolio losses and required capital under correlated risk scenarios to support underwriting recommendations
- Designed coverage terms including deductibles, limits, triggers and exclusions to reflect portfolio risk profiles and accumulation risk.

Parametric Flood Insurance Product Innovation | Insurance Innovation Competition, UNSW | 2026

- Co-developed a parametric flood insurance product for Australian households, combining rainfall and river-level triggers with three-tier payouts to support faster recovery
- Analysed historical rainfall, river-gauge and flood-event data across NSW and Queensland to identify and calibrate regional payout triggers, and evaluated their reliability, basis risk and customer outcomes
- Designed a monitoring and recalibration framework to assess payout adequacy, false-positive and false-negative trigger performance, affordability and portfolio concentration, with scope for future multi-peril expansion

Freely Travel Insurance: Conversion Analysis | Product Analytics Project, UNSW | 2025

- Performed extensive data cleaning, analysed quote-to-purchase conversion, identifying key drivers of conversion and drop-off
- Co-developed an interactive dashboard presenting predicted conversion rates and translating key drivers into product and customer insights; the team report was selected as an outstanding exemplar for the cohort

PROFESSIONAL EXPERIENCE

Insurance Sales Director Assistant Intern | Xiamen Ping An Insurance | July 2022

- Supported policy issuance, sales tracking and client communications
- Compared insurance product structures and coverage against customer needs

Web Designer | Xiamen OUHUA | Jul – Aug 2024

- Built and maintained the company website, using visitor data to improve content, navigation and user experience

ADDITIONAL COMMERCIAL EXPERIENCE

Reverse Garbage: Participation & Market Analysis | Social Enterprise Consulting Project, UNSW | 2025

- Investigated barriers to participation using market research and stakeholder insights, identifying key factors limiting customer engagement and uptake
- Translated findings into practical recommendations to strengthen participation and customer value; the team project was awarded First Prize

SELECTED ANALYTICAL PROJECTS

Financial Time Series & Dependence Risk Analysis | Individual Actuarial Project | R | 2026

- Analysed approximately ten years of NVIDIA and AMD equity data using time-series and copula models to assess dependence and tail behaviour
- Estimated portfolio VaR and Expected Shortfall, compared model results with empirical risk measures, and identified asymmetric downside dependence

Airbnb Predictive Modelling | Individual Machine Learning Project | R | 2025

- Built and evaluated predictive models through data cleaning, feature engineering, model comparison and out-of-sample validation to identify key drivers of the target outcome.
- Communicated model performance and practical implications in a concise analytical report selected as a high-quality exemplar for the cohort

TECHNICAL SKILLS

Analytics & Programming: R, Python, Excel, data visualisation, machine learning

Actuarial & Risk: Insurance pricing, risk modelling, reserving, VaR and Expected Shortfall, copulas, time-series analysis

Insurance & Commercial: Insurance product analysis, claims and exposure analysis, customer analytics, market research

Languages: Mandarin (native), English (professional working proficiency)